

# NanoGT Results: Cross-Disease GT Precedent Matching

**Algorithm:** 6-dimension scoring (packaging fit, tissue tropism, protein class, pathway similarity, inheritance compatibility, approval precedent). Max score = 10.

## Summary Table

| Type       | Disease                                        | ORPHA      | Gene    | CDS (bp) | #1 Precedent | Score  | Confidence |
|------------|------------------------------------------------|------------|---------|----------|--------------|--------|------------|
| Validation | Spinal Muscular Atrophy                        | ORPHA:70   | SMN1    | 882      | OAV101-IT    | 9.5/10 | high       |
| Validation | Hemophilia B                                   | ORPHA:3069 | CB      | 1383     | DTX201       | 9.6/10 | high       |
| Discovery  | Kohlschütter-Tonz syndrome                     | ORPHA:1946 | GDI     | 861      | ABO-101      | 8.5/10 | high       |
| Discovery  | Mucopolidosis type IV                          | ORPHA:571  | COLN1   | 11740    | ABO-101      | 9.0/10 | high       |
| Discovery  | Alpha-mannosidosis                             | ORPHA:61   | MAN2B1  | 3033     | ABO-101      | 8.5/10 | high       |
| Discovery  | Maple syrup urine disease                      | ORPHA:513  | BCKDHA  | 1335     | OAV101-IT    | 8.0/10 | high       |
| Discovery  | Salla disease                                  | ORPHA:308  | SLC17A5 | 1485     | ABO-101      | 9.0/10 | high       |
| Discovery  | Fabry disease                                  | ORPHA:324  | GLA     | 681      | RGX-121      | 8.3/10 | high       |
| Discovery  | Mucopolysaccharidosis type IIIA (Sanfilippo A) | ORPHA:796  | SGSH    | 1506     | ABO-101      | 9.0/10 | high       |
| Discovery  | Crigler-Najjar syndrome type I                 | ORPHA:1066 | UGT1A1  | 1599     | CPCB-RPE1    | 6.7/10 | medium     |

## Validation Results (positive controls)

### Spinal Muscular Atrophy (ORPHA:70)

**Gene:** SMN1 | **CDS:** 882 bp | **Inheritance:** Autosomal recessive | **Tissues:** CNS, muscle

1. **OAV101-IT** (AAV9) — 9.5/10 [high]
2. **Zolgensma** (AAV9) — 9.5/10 [high]
3. **AT132** (AAV8) — 9.0/10 [high]
4. **SRP-9001** (AAV9) — 8.2/10 [high]
5. **DTX301** (AAV8) — 6.8/10 [medium]

## Hemophilia B (ORPHA:306)

**Gene:** F9 | **CDS:** 1383 bp | **Inheritance:** X-linked recessive | **Tissues:** liver

1. **DTX201** (AAV8) — 9.6/10 [high]
  2. **Hemgenix** (AAV5) — 9.5/10 [high]
  3. **Roctavian** (AAV5) — 9.5/10 [high]
  4. **SPK-8011** (AAVrh10) — 9.3/10 [high]
  5. **RGX-121** (AAV9) — 8.3/10 [high]
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## Discovery Results (no approved GT)

### Kohlschutter-Tonz syndrome (ORPHA:1946)

**Gene:** ROGDI | **CDS:** 861 bp | **Inheritance:** Autosomal recessive | **Tissues:** CNS

1. **ABO-101** (AAV9) — 8.5/10 [high]
2. **RGX-121** (AAV9) — 8.5/10 [high]
3. **OAV101-IT** (AAV9) — 8.0/10 [high]
4. **Zolgensma** (AAV9) — 8.0/10 [high]
5. **DTX201** (AAV8) — 7.3/10 [medium]

### Mucopolidosis type IV (ORPHA:578)

**Gene:** MCOLN1 | **CDS:** 1740 bp | **Inheritance:** Autosomal recessive | **Tissues:** CNS, retina

1. **ABO-101** (AAV9) — 9.0/10 [high]
2. **RGX-121** (AAV9) — 9.0/10 [high]
3. **CPCB-RPE1** (AAV8) — 7.2/10 [medium]
4. **AVR-RD-01** (LV) — 7.0/10 [medium]
5. **OAV101-IT** (AAV9) — 7.0/10 [medium]

### Alpha-mannosidosis (ORPHA:61)

**Gene:** MAN2B1 | **CDS:** 3033 bp | **Inheritance:** Autosomal recessive | **Tissues:** CNS, liver

1. **ABO-101** (AAV9) — 8.5/10 [high]
2. **RGX-121** (AAV9) — 8.5/10 [high]
3. **ST-920** (AAV2/6) — 7.7/10 [high]
4. **DTX201** (AAV8) — 6.8/10 [medium]
5. **Hemgenix** (AAV5) — 6.7/10 [medium]

### Maple syrup urine disease (ORPHA:511)

**Gene:** BCKDHA | **CDS:** 1335 bp | **Inheritance:** Autosomal recessive | **Tissues:** liver, CNS

1. **OAV101-IT** (AAV9) — 8.0/10 [high]
2. **Zolgensma** (AAV9) — 8.0/10 [high]
3. **Luxturna** (AAV2) — 7.8/10 [high]
4. **DTX301** (AAV8) — 7.3/10 [medium]
5. **BMN 307** (AAV5) — 7.1/10 [medium]

### Salla disease (ORPHA:309)

**Gene:** SLC17A5 | **CDS:** 1485 bp | **Inheritance:** Autosomal recessive | **Tissues:** CNS

1. **ABO-101** (AAV9) — 9.0/10 [high]
2. **RGX-121** (AAV9) — 9.0/10 [high]
3. **AVR-RD-01** (LV) — 7.0/10 [medium]
4. **OAV101-IT** (AAV9) — 7.0/10 [medium]
5. **ST-920** (AAV2/6) — 7.0/10 [medium]

### Fabry disease (ORPHA:324)

**Gene:** GLA | **CDS:** 681 bp | **Inheritance:** X-linked dominant | **Tissues:** liver, kidney, heart, CNS

1. **RGX-121** (AAV9) — 8.3/10 [high]
2. **ABO-101** (AAV9) — 7.7/10 [high]
3. **SRP-9001** (AAV9) — 7.5/10 [high]
4. **ST-920** (AAV2/6) — 7.5/10 [high]
5. **CPCB-RPE1** (AAV8) — 6.9/10 [medium]

### Mucopolysaccharidosis type IIIA (Sanfilippo A) (ORPHA:79269)

**Gene:** SGSH | **CDS:** 1506 bp | **Inheritance:** Autosomal recessive | **Tissues:** CNS

1. **ABO-101** (AAV9) — 9.0/10 [high]
2. **RGX-121** (AAV9) — 9.0/10 [high]
3. **AVR-RD-01** (LV) — 7.0/10 [medium]
4. **OAV101-IT** (AAV9) — 7.0/10 [medium]
5. **ST-920** (AAV2/6) — 7.0/10 [medium]

### Crigler-Najjar syndrome type I (ORPHA:1060)

**Gene:** UGT1A1 | **CDS:** 1599 bp | **Inheritance:** Autosomal recessive | **Tissues:** liver

1. **CPCB-RPE1** (AAV8) — 6.7/10 [medium]
2. **SRP-9001** (AAV9) — 6.7/10 [medium]
3. **RGX-121** (AAV9) — 6.5/10 [medium]
4. **DTX201** (AAV8) — 6.3/10 [medium]
5. **DTX301** (AAV8) — 6.3/10 [medium]